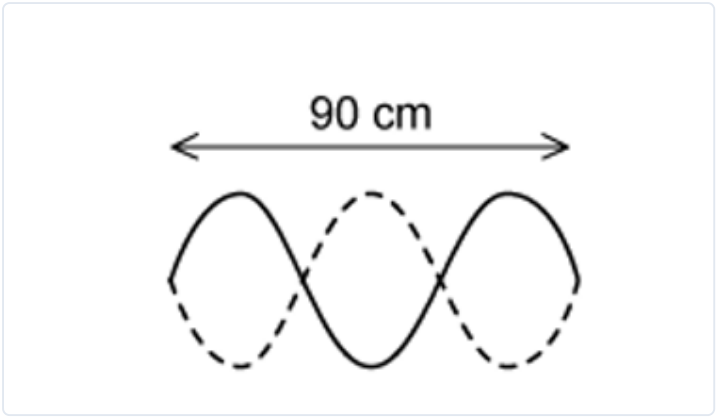
CIE AS PHYSICS (9702) • QUESTION PAPER

8.1 Stationary Waves

Syllabus Section: 8. Superposition **Paper Type:** Multiple Choice (Paper 1) **Total:** 31 Questions (31 Marks)

Question 1 Easy [1 mark]

A stationary wave shown in the diagram was created on a stretched string, it had two instants of maximum vertical displacement.



The frequency of the wave is 13 Hz.

What is the speed of the wave?

A $3.9\ m\ s^{-1}$

B $7.8\ m\ s^{-1}$

C $390\ m\ s^{-1}$

D $780\ m\ s^{-1}$

Question 2 Easy [1 mark]

The table shows some statements about stationary and progressive waves.

Which row is correct ?

	Progressive wave	Stationary wave
A	particles one wavelength apart vibrate in phase	particles one wavelength apart vibrate in phase
B	particles vibrate in phase with their immediate neighbours	particles in adjacent loops vibrate in antiphase
C	energy is transferred along the wave	energy is transferred along the wave
D	all particles vibrate with the same amplitude	all particles vibrate with the same amplitude

A

B

C

D

Question 3 Easy [1 mark]

When a stationary wave is produced in a tube, which of the following statements describes the wave speed?

A it is the speed of a particle at a node

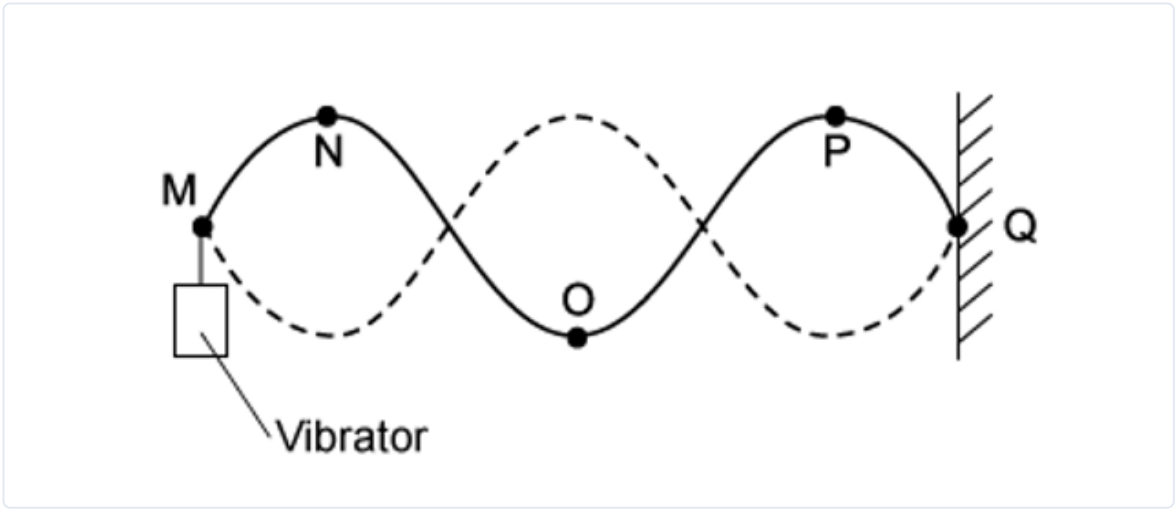
B it is the speed of one of the progressive waves that are producing the stationary wave

C it is the speed at which energy is transferred from one antinode to an adjacent antinode

D it is the distance between two adjacent nodes divided by the period of the wave

Question 4 Easy [1 mark]

A stretched string is set up to demonstrate a stationary wave between two points M and Q.



Which of the following statements are correct?

A the distance between M and Q is three wavelengths

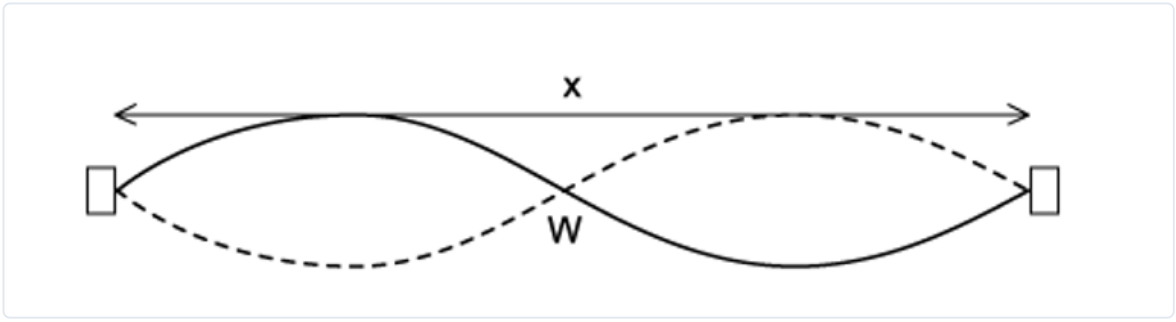
B the wave shown has the lowest possible frequency

C points N and P vibrate in phase

D point O is at a node

Question 5 Easy [1 mark]

A stretched string is used to demonstrate a stationary wave, as shown in the diagram.



Which row in the table correctly describes the length of x and the name of W?

	Length x	Point W
A	two wavelengths	node
B	one wavelength	node
C	two wavelengths	antinode
D	one wavelength	antinode

A

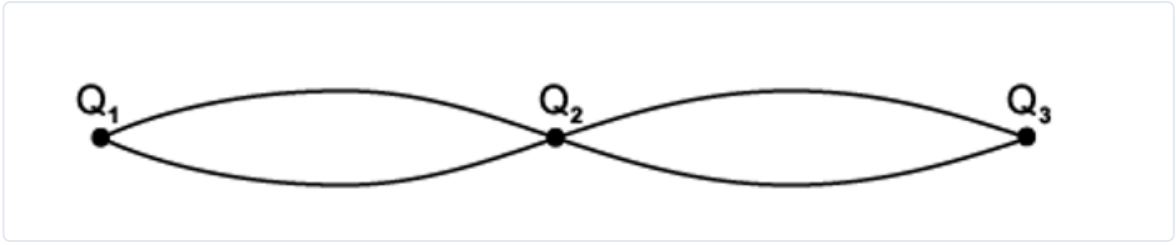
B

C

D

Question 6 Easy [1 mark]

A stationary wave on a string is shown in the diagram below. The standing wave has three nodes Q_1, Q_2 and Q_3



Which of the following statements are correct ?

A all points on the string vibrate in phase

B all points on the string vibrate with the same amplitude

C points equidistant from Q_2 vibrate with the same frequency and the same amplitude

D points equidistant from Q_2 vibrate with the same frequency and in phase

Question 7 Easy [1 mark]

A student formed a stationary wave in a tube by blowing over the top as shown in the diagram



Which of the following statements are correct ?

A the stationary wave in the cylinder is polarised

B the stationary wave will have an antinode at the bottom of the cylinder

C the fundamental frequency of the stationary wave decreases when some water is added to the cylinder

D the stationary wave in the cylinder is caused by the superposition of two waves moving in opposite directions

Question 8 Easy [1 mark]

A stationary wave is formed when two waves travel in opposite directions through the same space.

Which of the following statements must also be correct?

A the waves must have equal speed, frequency and wavelength

B the waves must have equal frequency, but a different speed and wavelength

C the waves must have equal wavelength, but a different speed and frequency

D the waves must have equal speed, but a different wavelength and frequency

Question 9 Easy [1 mark]

A student sets up a sound wave in a long tube, and the tube is closed at one end. The tube length is changed until the sound from the tube is loudest.

Which of the following statements describe the nature of the wave in the tube?

A transverse and stationary

B transverse and progressive

C longitudinal and stationary

D longitudinal and progressive

Question 10 Easy [1 mark]

A student wants to demonstrate a stationary wave.

What could they use to produce this?

A passing water waves through a narrow slit

B passing monochromatic light through a double slit

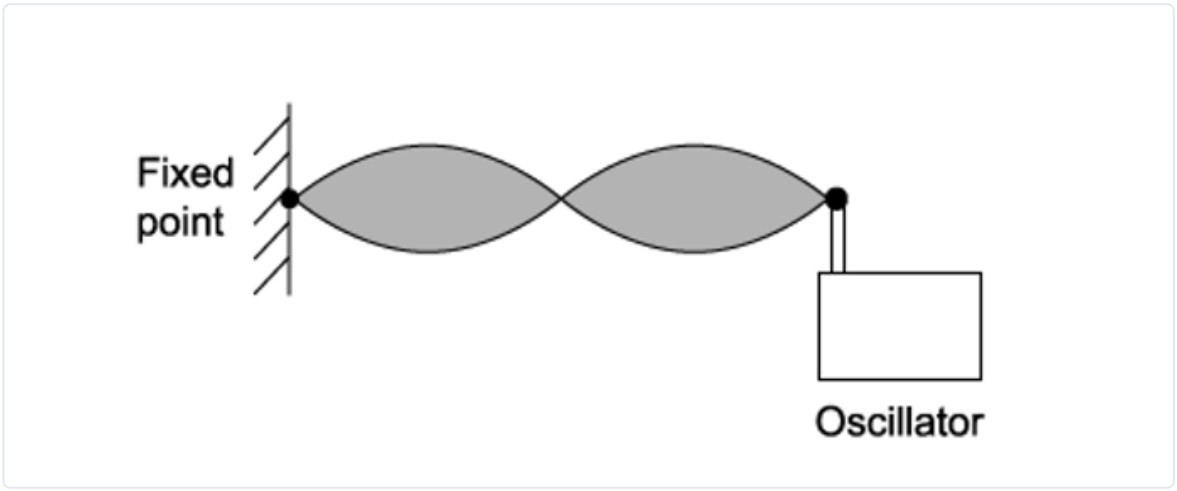
C making a loud sound near a mountain

D blowing air over the top of an empty bottle

Question 11 Medium [1 mark]

A student was investigating the speed of a transverse wave on a stretched spring. They found that by adjusting the tension on the spring, they were able to change the speed.

The wave pattern produced is shown in the diagram the frequency of the oscillator was set to 650 Hz



What needs to be changed to maintain the same wave pattern when the frequency is increased to 750 Hz?

A increase the speed of the wave on the string

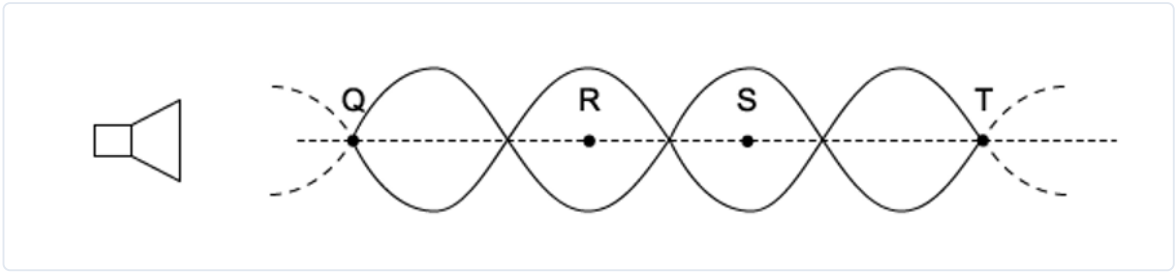
B increase the wavelength of the wave on the string

C decrease the wavelength of the wave on the string

D decrease the speed of the wave on the string

Question 12 Medium [1 mark]

The pattern produced when two waves are produced by superposition of sound waves from a loudspeaker reflected by a metal sheet.



Q, R, S and T are four points on the line through the centre of these waves.

Which of the following statements is correct?

A a node is a quarter of a wavelength from an adjacent antinode

B the stationary waves oscillate at right angles to the line QT

C an antinode is formed at the surface of the metal sheet

D the oscillations at R are in phase with those at S

Question 13 Medium [1 mark]

Headphones that have noise reduction can cancel out external noise by producing their own waves.

They are fitted with a microphone that detects the external sound frequency. The loudspeaker then produces a wave with the same frequency but different phase.

What would the phase difference needed to cancel out the external sound?

A 360°

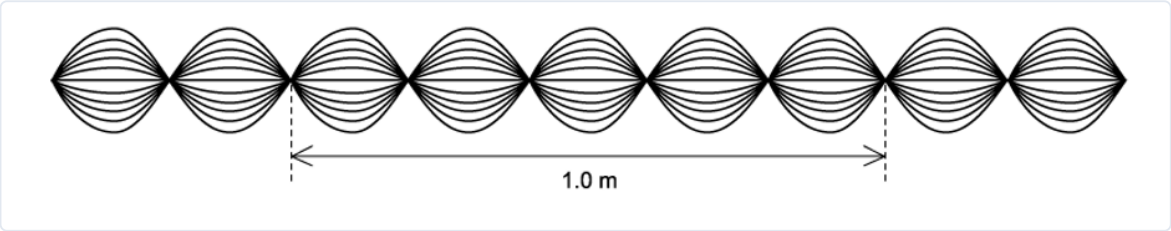
B 270°

C 180°

D 90°

Question 14 Medium [1 mark]

The diagram below shows a wave pattern over time



What is the correct description of the wave?

A the wave is transverse, has a wavelength of 40 cm and is stationary

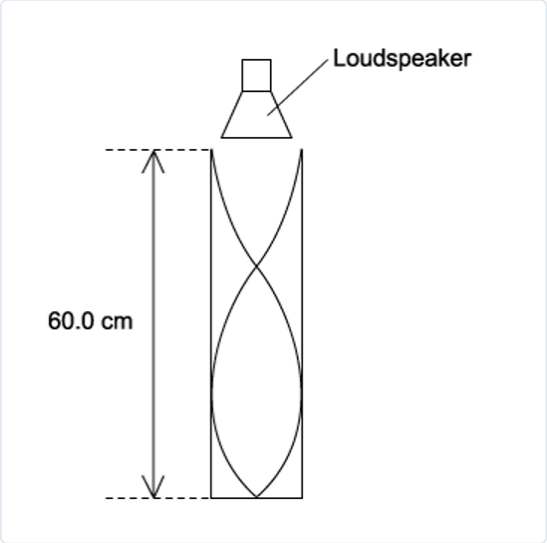
B the wave is transverse, has a wavelength of 40 cm and is progressive

C the wave is transverse, has a wavelength of 20 cm and is stationary

D the wave is longitudinal, has a wavelength of 20 cm and is stationary

Question 15 Medium [1 mark]

The diagram below shows sound wave produced in a column by a loudspeaker. The speed of sound in air is 330 m s^{-1}



What is the frequency of the wave?

A 1650 Hz

B 830 Hz

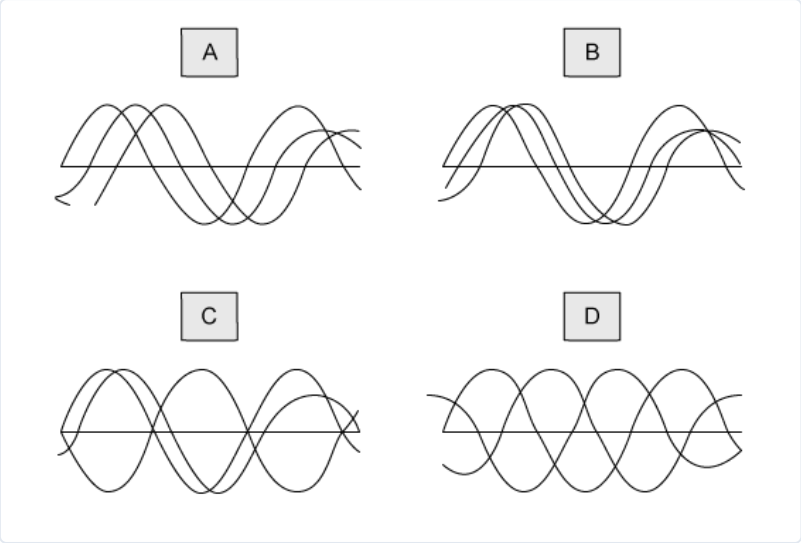
C 550 Hz

D 413 Hz

Question 16 Medium [1 mark]

Each of the diagrams below show three waves; each of the waves have the same amplitude and frequency but have different phases.

The waves are added together to give a resultant wave. In which case would the resultant wave be zero ?



A

B

C

D

Question 17 Medium [1 mark]

A stationary sound wave produced in a tube has a series of nodes. The distance between the first node and the sixth node is 30 cm

What is the wavelength of the wave ?

A 12.0 cm

B 10.0 cm

C 6.0 cm

D 5.0 cm

Question 18 Medium [1 mark]

A stationary sound wave produced in a musical instrument, this instrument has the pipes open at one end and closed at the other.

If the length of the pipe was 10 m what it is the lowest frequency of sound when the speed of sound is 320 m s^{-1}?

A 32 Hz

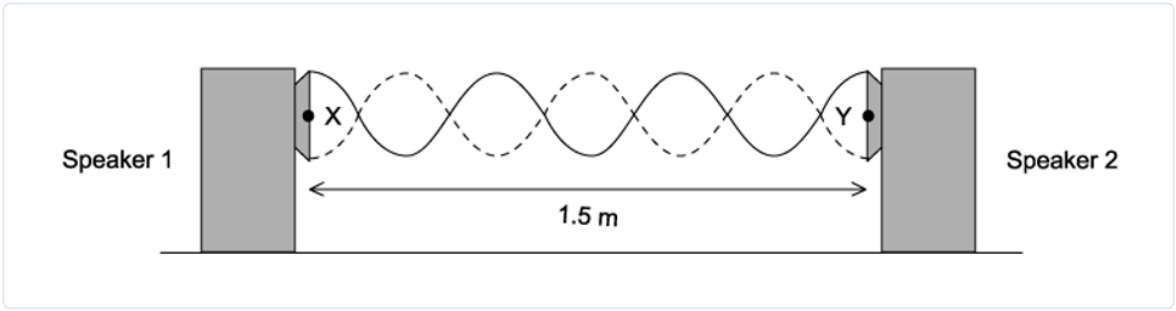
B 16 Hz

C 8 Hz

D 4 Hz

Question 19 Medium [1 mark]

Two loudspeakers emitting sound of the same frequency produces a stationary wave. This is shown in the diagram



The microphone is moved between X and Y for a distance of 1.5 m. There were six nodes and seven antinodes observed.

What is the wavelength of the sound?

A 0.21 m

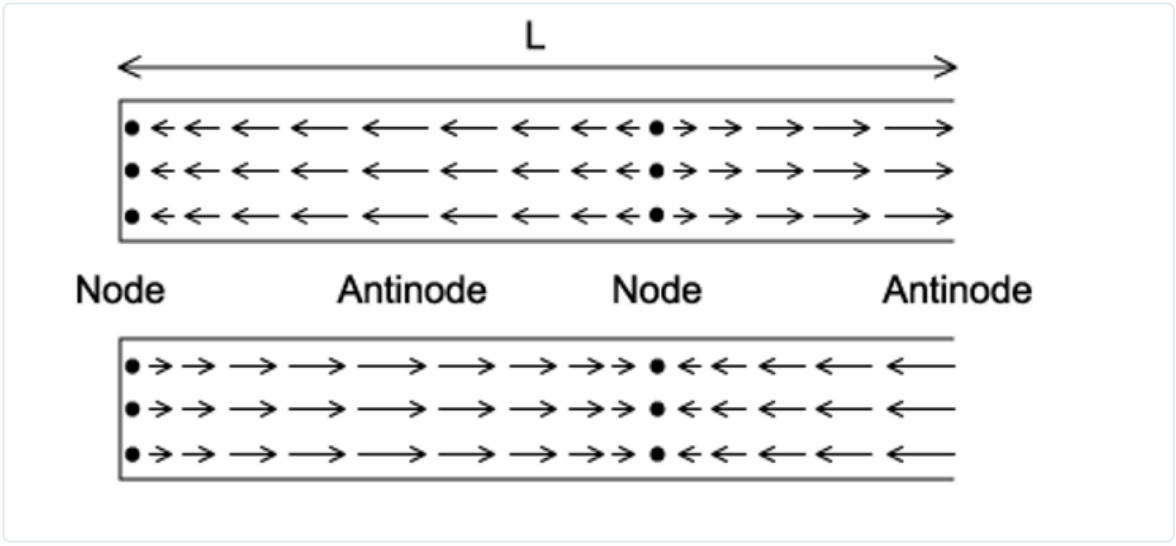
B 0.25 m

C 0.43 m

D 0.50 m

Question 20 Medium [1 mark]

A stationary wave was formed in a column. The diagram below shows the movement of the air particles.



The first diagram shows the displacement and the second is the displacement half a cycle later.

Which position will the pressure change by the most significant amount, and what is the length, L of the column in terms of wavelength, λ?

	Length L	Maximum pressure change
A	$\frac{3}{4}\lambda$	antinode
B	$\frac{3}{4}\lambda$	node
C	$\frac{3}{2}\lambda$	antinode
D	$\frac{3}{2}\lambda$	node

A

B

C

D

Question 21 Medium [1 mark]

A standing wave is formed in a closed pipe. The distance between the first node and the ninth node is 18.0 cm.

What is the wavelength of the wave?

A 4.5 cm

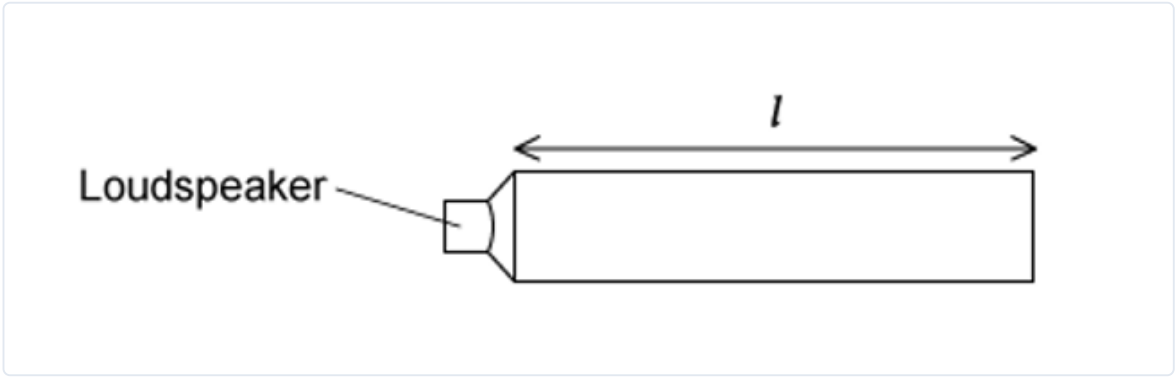
B 9.0 cm

C 12.0 cm

D 24.0 cm

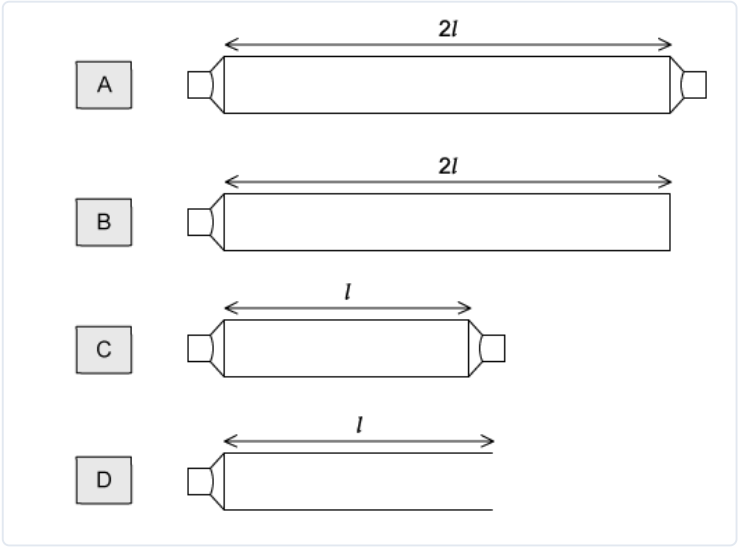
Question 22 Hard [1 mark]

A demonstration of a standing wave is set up using a loudspeaker. The loudspeaker is emitting sound with a frequency *f* and is placed at one end of the pipe with length *l*. The pipe is closed at the other end. This is shown in the diagram below.



Different pipes were set up with either one or two loudspeakers of frequency *f*. The pairs of loudspeakers were in phase with each other.

Which pipe would contain a standing wave ?



A

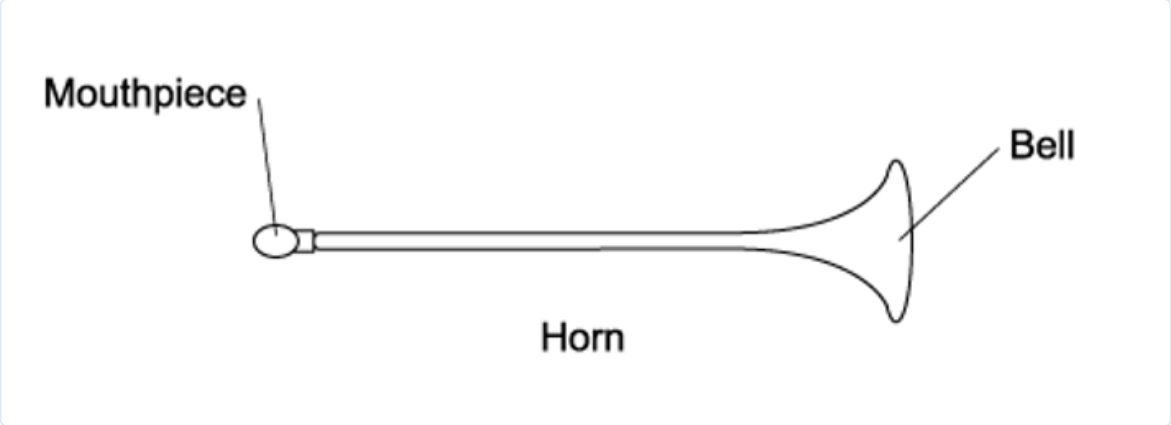
B

C

D

Question 23 Hard [1 mark]

When a wind instrument is used to create a note, a stationary wave is produced in the column. An example of this is the horn.



For the range of notes produced by the horn, a node is formed at the mouthpiece and an antinode is formed at the bell. The frequency of the lowest note is 75 Hz.

What would be the frequencies of the next two higher notes for this air column?

	First higher note / Hz	Second higher note / Hz
A	225	375
B	150	300
C	150	225
D	113	150

A

B

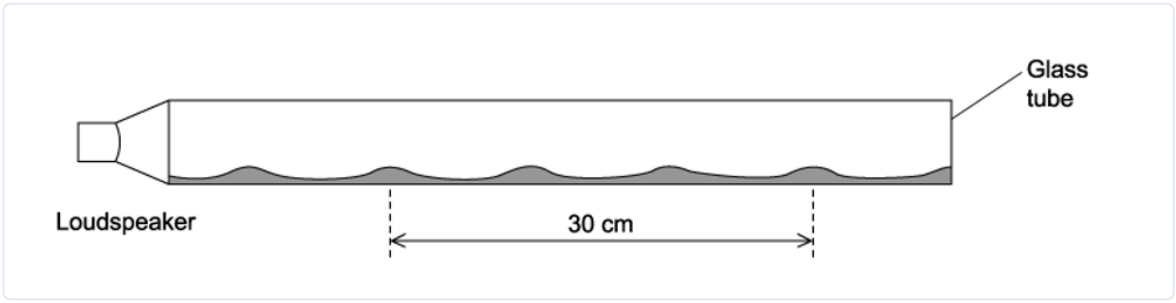
C

D

Question 24 Hard [1 mark]

A glass tube closed at one end, with a layer of fine powder laid inside as shown in the diagram, was set up. Sound was emitted from the loudspeaker and placed near the open end of the tube.

The frequency of the sound wave was changed, at one frequency a stationary wave was formed, and the powder formed 4 heaps. The distance between the four heaps of powder was 30 cm.



The speed of sound in the tube is 330 m s^{-1}

What is the frequency of the sound?

A 1650 Hz

B 2200 Hz

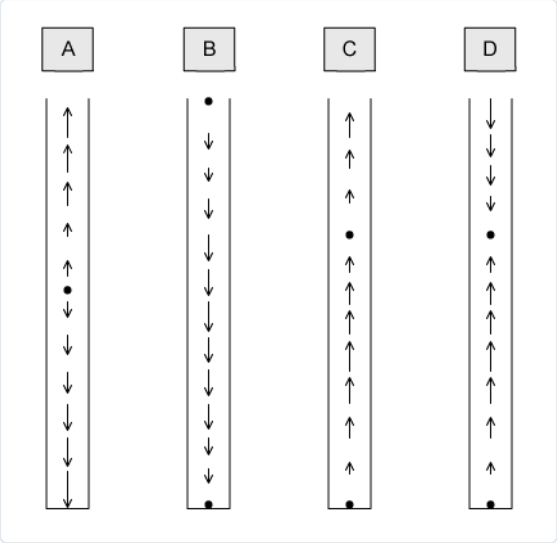
C 3300 Hz

D 6600 Hz

Question 25 Hard [1 mark]

A pipe is used to set up a stationary longitudinal wave. The length of each arrow in the diagrams below shows the motion of the air molecules. The arrowheads show the direction of the motion of the particle.

Which diagram shows a stationary wave with two nodes and two antinodes?



A

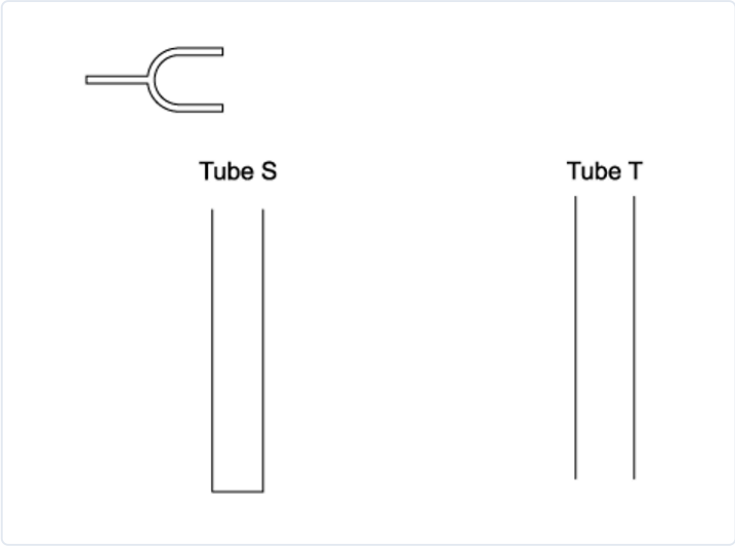
B

C

D

Question 26 Hard [1 mark]

A student was investigating a tuning fork over two tubes. The tube were identical in length and size except tube S is closed at one end, and tube T is open at the lower end.



The tuning fork was placed above tube S and caused a resonance of the air at frequency *f*. No resonance is found at any lower frequency than *f* with tube S.

Which tuning fork will produce resonance when placed above tube T?

A a fork of frequency $2f$

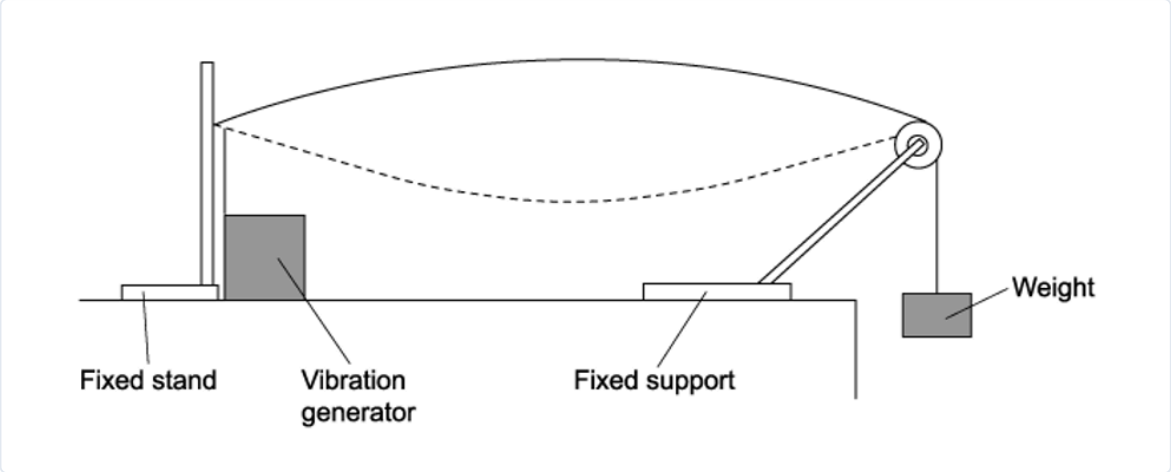
B a fork of frequency $\frac{3f}{2}$

C a fork of frequency $\frac{2f}{3}$

D a fork of frequency $\frac{f}{2}$

Question 27 Hard [1 mark]

A steel wire was set up clamped at one end and placed under tension by a mass hung over a pully; this is shown in the diagram.



A vibration generator is attached to the wire nearest to the clamped end. A stationary wave with one loop is produced. The frequency of the generator is *f*.

Which frequency should be used to produce two loops?

A $2 \quad f$

B $4 \quad f$

C $\frac{f}{2}$

D $\frac{f}{4}$

Question 28 Hard [1 mark]

An investigation to sound waves reflected by a wall was carried out. The frequency *f* of the wave was adjusted until a stationary wave is formed with an antinode with a distance *x* closest to the wall.

Which expression gives *f* in terms of *x* and the speed of sound *c* ?

A $f = \frac{c}{2x}$

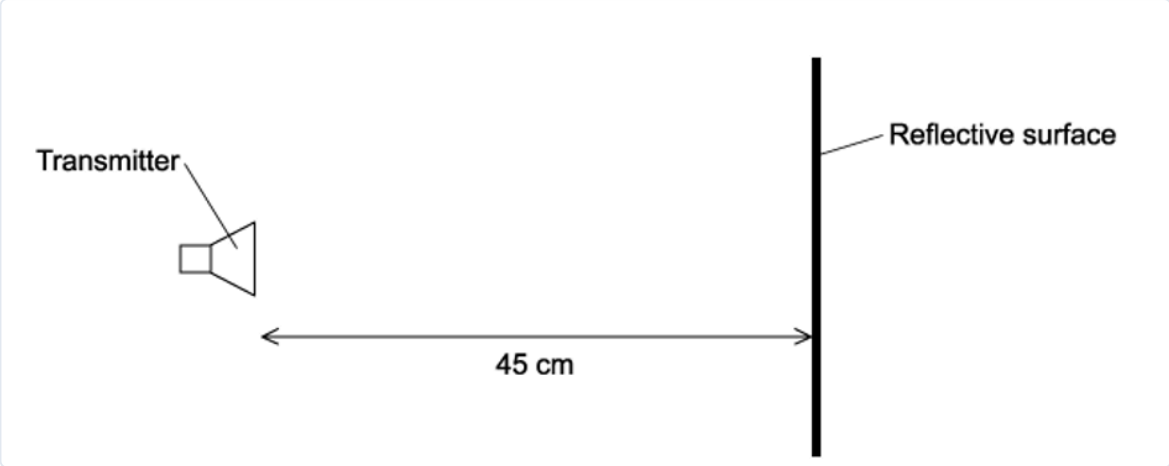
B $f = \frac{c}{4x}$

C $f = \frac{2c}{x}$

D $f = \frac{4c}{x}$

Question 29 Hard [1 mark]

A student was investigating the reflection of electromagnetic waves. They set up a transmitter 45 cm from a reflective surface, as shown in the diagram.



The waves had a frequency of 1.00 GHz. A stationary wave was produced with a node at the transmitter and a node at the surface.

How many antinodes are there in the space between the surface and the transmitter?

A 4

B 3

C 2

D 1

Question 30 Hard [1 mark]

Stationary sound waves can be produced in an organ pipe of length L when both ends are open, and this creates a note from an organ.

The speed of sound in the air column is v.

What is the lowest (fundamental) frequency of the note produced in the pipe?

A $\frac{v}{4\mathrm{L}}$

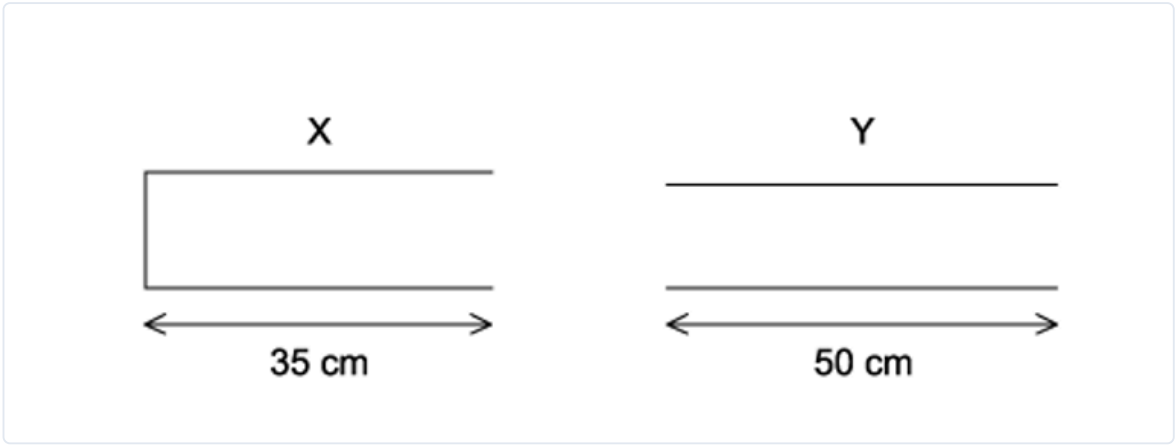
B $\frac{v}{2\mathrm{L}}$

C $\frac{v}{\mathrm{L}}$

D $\frac{2v}{4\mathrm{L}}$

Question 31 Hard [1 mark]

A student was investigating the creation of stationary waves. They set up travelling waves with a wavelength of 20 cm in a closed pipe X and an open pipe Y. The lengths of the pipes are shown in the diagram.



In which pipe did the student observe stationary waves?

A X only

B X and Y

C Y only

D neither X or Y